AI IN SPACE AND ORBITAL OPERATIONS - VOLUME 1

The AI Space Strategy Playbook

Where space AI pays, where it should never go, and how to phase autonomy without losing a mission.



Gates SS0-SS5 | Eight templates | 20 golden rules

A modelled manoeuvre is not a flown mission. Space AI is licensed per spacecraft, per orbit, per pass and per failure mode - never in general, and never on a simulation.

CHAPTER 1 - THE DECISION BRIEF

A mission you cannot recall is a decision you cannot rush

Every other domain in this library can roll back. A database restores, a campaign pauses, a robot stops. A spacecraft on a wrong burn, a constellation commanded into a bad configuration, or a lander committed to a bad descent cannot be recalled, patched in person, or apologised to. That single physical fact, not model quality, is the strategy.

This volume is for executives who fund space AI, mission directors who sign for it, and operators who will live with it at 3 a.m. on an anomalous pass. It separates three questions that vendors habitually merge: can the model do it in simulation, may it do it on this spacecraft, and who answers when it does it wrong in flight?

The brief in one page

- **Start where data is dense and actions reverse:** telemetry triage, pass planning, image screening, anomaly detection.
- **Phase autonomy by consequence:** advise, then propose, then act inside a signed envelope, then act across a fleet - each step separately evidenced.
- **Price the downside honestly:** one lost spacecraft can erase a decade of operator hours saved.
- **Keep a human authority that outranks the model** on every irreversible action, with a tested halt.

The licensing rule: **per spacecraft, per orbit, per pass, per failure mode** - never in general, never on a simulation.

EXECUTIVE LENS

Fund the reversible work first; make irreversible autonomy earn its way in.

MISSION LENS

Name the failure mode you fear most before you name the use case.

OPERATIONS LENS

A demo in the simulator is an application, not a licence.

Five rungs separate a model output from a mission fact

Rung	Name	What it proves	What it does not
1	Modelled	The system produces the output in offline data	Anything about flight
2	Simulated	It holds in a high-fidelity simulator and hardware-in-the-loop	Behaviour in the real space environment
3	Authorised	A named human has signed an envelope: bounds, aborts, expiry	Performance outside that envelope
4	Flown	It has operated in flight, inside the envelope, with telemetry to prove it	Transfer to another bus, orbit or regime
5	Transferable	Re-evidenced on the new spacecraft, orbit or mission class	Nothing is transferable by assertion

Most vendor claims live at rung 1 or 2 and are sold in rung 4 language. The discipline of this series is to ask, of every claim, which rung it sits on, what evidence put it there, and when that evidence goes stale. Radiation, ageing, thermal drift and a changing debris environment all erode evidence in orbit; a licence that was true at launch is a hypothesis by year three.

Claims also decay by change. A software upload, a new sensor mode, a different ground station or a neighbouring constellation's manoeuvre can each drop a system a rung. The register in Chapter 19 exists to catch that drop before a pass does.

EXECUTIVE LENS

Ask for the rung before the roadmap.

MISSION LENS

A rung 2 demo priced as rung 4 is the commonest buying error in space AI.

OPERATIONS LENS

Evidence has a half-life in orbit; schedule its renewal.

Grade every stage before you automate any of it

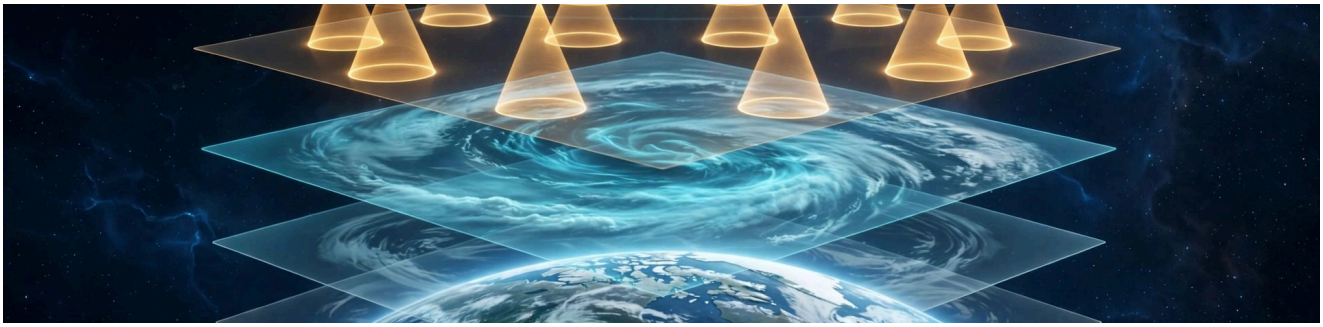


Figure 1 - The operational stack as layered time planes: orbit, debris, weather, coverage and ground, each graded separately.

Stage	AI today	Grade	First licence
Mission design and trade studies	Design-space search, trade-off ranking	Strong assist	Advisory
Flight dynamics and orbit determination	Estimation, manoeuvre planning	Strong, bounded	Propose, human signs
Pass and resource planning	Scheduling under constraints	Strong	Act, reversible
Telemetry and anomaly triage	Pattern detection at fleet scale	Strongest case	Act, alerting only
Payload data screening	Onboard and ground triage	Strong	Act, with sampling audit
Conjunction screening	Risk ranking, manoeuvre options	Assist only	Propose, human decides
Autonomous manoeuvre execution	Bounded burns inside an envelope	Emerging	Narrow, per spacecraft
Rendezvous and servicing	Relative navigation support	Emerging, high stakes	Hold points mandatory
Entry, descent and landing	Real-time guidance support	Special case	Pre-committed, no learning in flight
End-of-life and disposal	Planning and passivation checks	Assist	Human-signed, regulator-visible

The map is the strategy. Stages differ by data density, reversibility and time pressure, and a vendor excellent at telemetry triage has earned nothing at conjunction decisions. Licence stage by stage, and let the strongest stages fund the careful exploration of the hardest ones.

EXECUTIVE LENS

The portfolio, not the model, is the unit of strategy.

MISSION LENS

Conjunction and disposal stay human-decided longest, by design.

OPERATIONS LENS

Fund autonomy from the savings of the safe stages.

The value is in passes, pictures and patterns - in that order

Passes. A constellation lives or dies on contact time. AI scheduling that packs ground-station passes, resolves conflicts across a fleet and re-plans around a failed contact converts directly into data returned and commands delivered. The action is reversible - a bad plan can be re-planned next pass - which is exactly why it licenses early.

Pictures. Earth-observation and inspection payloads produce far more data than downlink allows. Onboard screening that downlinks the cloud-free, the changed and the anomalous first raises the value of every megabit. The failure mode is a missed event, so the licence carries a sampling audit: a fixed share of rejected data is downlinked anyway and scored.

Patterns. Fleets generate telemetry no human team can watch channel by channel. Anomaly detection that has learned a fleet's normal catches the slow battery fade and the odd reaction-wheel signature weeks earlier than thresholds do. It alerts; humans diagnose. That division is the licence.

Notice what is absent: none of the three highest-value uses touches the spacecraft's trajectory. Value and irreversibility are, usefully, almost inversely related in space AI. Strategy exploits that instead of fighting it.

EXECUTIVE LENS

Sequence by reversibility, not by glamour.

MISSION LENS

Downlink is the scarcest resource in most missions; optimise for it.

OPERATIONS LENS

An alerting licence is not a lesser licence - it is the right one.

Some envelopes stay closed regardless of model scores

- **Unbounded trajectory change.** No system may design and execute an unreviewed burn that takes a spacecraft outside its authorised orbital envelope.
- **Weapons-adjacent and interference functions.** Deliberate jamming, blinding, grappling or proximity operations against another operator's asset without that operator's agreement sit outside any commercial licence in this series.
- **Safety-of-life decisions without a human in authority** on crewed missions: life support, abort and docking decisions may be advised, never owned, by a model.
- **Learning in flight during critical phases.** A model that updates itself during entry, descent, landing or rendezvous is unverifiable by construction; critical phases fly frozen, validated configurations.
- **Disposal shortcuts.** AI may plan passivation and disposal; it may not trade disposal compliance against propellant or revenue.
- **Silent authority expansion.** Any system that widens its own envelope, thresholds or permissions without a signed change is grounded on discovery, whatever it did meanwhile.

These refusals are standing, not situational. They are written into procurement, so a vendor pitch that requires one of them ends at the pitch, and into operations, so schedule pressure on the night cannot reopen them.

EXECUTIVE LENS

Refusals protect the programme from its own busiest night.

MISSION LENS

Frozen configurations for critical phases is a design rule, not a preference.

OPERATIONS LENS

If a use needs a refusal lifted, it needs a new programme, not an exception.

Four levels of authority, each earned separately

Level	Authority	Evidence required
A1 Advise	System informs; human acts	Accuracy on historical and simulated data
A2 Propose	System drafts the command or plan; human signs each one	Acceptance and override rates, error taxonomy
A3 Act in envelope	System acts inside signed bounds; human supervises, can halt	Shadow-mode flight evidence, abort drills, envelope testing
A4 Act across fleet	Coordinated action across spacecraft	A3 per spacecraft, plus interaction and cascade testing

Levels attach to a function on a spacecraft, not to a product. The same vendor system can hold A3 for pass planning and A1 for conjunction response on the same fleet, and that asymmetry is healthy.

Promotion is a gate event with evidence, a named signer and an expiry. Demotion is automatic on defined triggers: an out-of-envelope action, a missed anomaly class, an unexplained telemetry gap. A system that cannot be demoted automatically will eventually be trusted past its evidence, which is how anomalies become losses.

EXECUTIVE LENS

Buy levels, not autonomy in the abstract.

MISSION LENS

Automatic demotion is the feature that makes promotion safe.

OPERATIONS LENS

Expiry forces the re-evidence conversation vendors avoid.

Design for the pass you miss and the hour you cannot reach it

Space operations run on orbital mechanics, not office hours. Contact may be minutes per pass and hours apart; light-time delay makes lunar operations deliberate and deep-space operations asynchronous by minutes to tens of minutes. Autonomy strategy is therefore partly a bet about communication: the further and rarer the contact, the more authority must travel with the spacecraft, and the tighter its envelope must be drawn in advance.

Three design consequences

- **Safe mode is a destination, not a failure.** Every autonomous function needs a tested path to a known safe state it can reach alone, and the licence counts safe-mode entries as data, not embarrassments.
- **Decisions carry deadlines.** A conjunction warning, a battery anomaly and a thermal limit each have a decision clock; the envelope states who or what decides when the clock runs out mid-gap.
- **Everything waits for telemetry.** No claim about an autonomous action is settled until telemetry confirms it, which may be a pass or a day later. Reporting discipline in Chapter 19 follows from this.

EXECUTIVE LENS

Distance buys autonomy and charges for it in envelope discipline.

MISSION LENS

Write the no-contact decision before the mission needs it.

OPERATIONS LENS

Safe-mode entries are evidence; hiding them is a culture failure.

Telemetry is the mission's memory; treat its gaps as findings

Space AI trains and runs on telemetry, tracking and payload data that are sparse, delayed, lossy and unlike any terrestrial dataset. Channels drop, values freeze, clocks drift, and the most interesting events are by definition the ones the training set contains least. Strategy therefore funds the data layer before the model layer: time synchronisation, provenance per channel, gap marking rather than silent interpolation, and an archive that preserves raw values alongside every derived product.

- **Raw is sacred.** Models may annotate telemetry; they may never overwrite it. Every alert links back to raw channels an engineer can inspect.
- **Gaps are labelled.** An interpolated channel presented as measured is a small fraud that compounds into bad training data.
- **Rare events are curated.** Every real anomaly, however embarrassing, enters a labelled library with its outcome. That library, over years, is worth more than any model bought.
- **Cross-mission data is negotiated, not assumed.** Another operator's telemetry, or a manufacturer's fleet data, comes with rights, formats and incentives that must be contracted explicitly.

EXECUTIVE LENS

The anomaly library is the moat; fund it like one.

MISSION LENS

A model trained on interpolated gaps learns a spacecraft that never existed.

OPERATIONS LENS

Provenance per channel, or no licence.

The simulator is where autonomy earns its application - and its limits

A high-fidelity simulator with hardware-in-the-loop is the only place a space AI can fail a thousand times cheaply. The strategy requirement is honesty about the seam: every simulator omits something - radiation effects, sensor degradation, a thruster's true plume, the other operator's manoeuvre - and the licence records which omissions matter for this function.

- **Twin fidelity is graded per phenomenon,** not claimed in general. A twin excellent for power may be fiction for thermal.
- **Monte Carlo with adversarial draws:** disperse initial conditions, inject faults, and include the cases the design team finds implausible.
- **Sim-to-flight reconciliation:** after every flown autonomous action, predicted and actual telemetry are compared and the delta is logged against the twin. Persistent deltas demote the twin, and with it the licences that rest on it.
- **The twin never signs.** Simulation evidence supports a human's authorisation; it does not replace it, however many runs pass.

EXECUTIVE LENS

Ask which phenomena the twin is wrong about; a good vendor has a list.

MISSION LENS

Reconciliation after flight is what keeps simulation honest.

OPERATIONS LENS

A million passing runs prove the model fits the simulator.

Most space AI value, and most of its risk, lives on the ground

Ground systems plan, command, receive and archive, and they are where AI deploys fastest because they patch easily and fail recoverably. They are also the attack surface and the error surface: a bad plan uploaded at one pass executes hours later, far from its author.

- **Command assurance:** AI-drafted command loads pass independent verification - syntax, bounds, interlocks and a second-system check against the flight rules - before uplink. The checker is deliberately dumber than the planner.
- **Pass discipline:** what can be uplinked in a pass, by whom, under what approval, is stated per mission phase; contingency passes have their own, shorter list.
- **Ground station diversity:** scheduling AI optimises across networks with different weather, politics and failure modes; concentration risk is reported, not just efficiency.
- **Patch windows:** ground AI updates fly under change control tied to the mission calendar, never mid-anomaly unless the anomaly is the software itself.

EXECUTIVE LENS

The independent checker is non-negotiable, and deliberately simple.

MISSION LENS

Efficiency that concentrates ground risk is not efficiency.

OPERATIONS LENS

Change control follows the mission calendar, not the sprint calendar.

Onboard AI works within watts, radiation and no second chances

Onboard intelligence operates under constraints terrestrial AI never meets: limited compute and power, radiation-induced faults, thermal extremes, and no possibility of a site visit. Strategy therefore favours small, verifiable onboard functions - screening, compression, fault detection, bounded control - over large general models, and treats every onboard update as a mission event.

- **Verify on the ground, freeze for flight:** candidate models complete ground validation and fly as fixed configurations with checksums and rollback images.
- **Watchdogs independent of the model:** hardware and simple-software monitors bound what any model output can command, and can cut it out of the loop.
- **Radiation realism:** single-event upsets are a design case, tested by fault injection, not a footnote.
- **Update cadence is a risk decision:** each upload states its rung evidence, its rollback, and the passes reserved to verify it.

EXECUTIVE LENS

Small and verifiable beats large and impressive at 550 kilometres.

MISSION LENS

The watchdog must not share the model's failure modes.

OPERATIONS LENS

An upload without a rehearsed rollback is a hope, not a plan.

The orbit is shared; your licence ends where another operator begins

Low Earth orbit is a commons filling with active satellites and tracked debris. Conjunction screening turns tracking data of varying quality into decisions under uncertainty: manoeuvre and spend propellant and mission life, or hold and accept a probability. AI earns its keep ranking, filtering false alarms and drafting options; the decision, and coordination with the other operator, stay human and documented.

- **Data quality is stated with every alert:** covariance, source and age, so a confident interface never launders uncertain astrometry.
- **Manoeuvre externalities are counted:** your avoidance burn changes everyone else's screening picture; coordination protocols are part of the licence.
- **Thresholds are policy, signed in advance,** not tuned under pressure during a high-interest event.
- **Debris creation is a hard boundary:** no autonomous action may foreseeably fragment a spacecraft or abandon a derelict outside disposal rules.

Volume 6 of this series treats navigation, timing and debris avoidance in full; this volume fixes the strategic line: assist strongly, decide humanly, coordinate always.

EXECUTIVE LENS

A probability is not a permission; thresholds are signed policy.

MISSION LENS

Your burn is someone else's conjunction. Coordinate.

OPERATIONS LENS

False-alarm fatigue is a safety metric, not an annoyance.

No link, no licence: autonomy must degrade as gracefully as it performs

Every space AI function sits on a link budget: spectrum allocations, ground weather, interference, and for deep space, the Deep Space Network's scarce schedule. Strategy maps each function to its link assumptions and tests the degraded cases: What does the system do on a missed pass, a corrupted command, a spoofed signal, a week of solar weather?

- **Degradation is designed, rehearsed and graded,** with a published ladder from full service to safe hold.
- **Store-and-forward honesty:** data delayed is data labelled delayed; dashboards never present stale telemetry as live.
- **Spectrum is licensed reality:** AI planning respects filings and coordination agreements as hard constraints, not optimisation hints.
- **Latency budgets decide placement:** functions that must act inside the light-time or the pass gap belong onboard, inside envelopes; everything else can afford the ground and its review.

EXECUTIVE LENS

Test the missed pass before the perfect one.

MISSION LENS

Stale data shown as live is an operations incident.

OPERATIONS LENS

Placement - ground or onboard - follows the latency budget.

Name the failure modes first; the model is the last line, not the first

Failure mode	First defence	AI's role
Wrong command or plan	Independent verification, interlocks	Detection, never sole author and checker
Missed anomaly	Thresholds and watchdogs in parallel	Earlier warning, graded on misses
False-alarm flood	Alert budgets per shift	Ranking, with fatigue measured
Out-of-envelope action	Hard bounds outside the model	None - bounds are not learned
Loss of link	Pre-signed contingency behaviour	Execution of the signed plan only
Cascade across fleet	Per-spacecraft authority limits	Co-ordination within caps

The safety case is a living document per function: hazards, defences, evidence, residual risk and the name of the person who accepted it. It is reviewed on every promotion, every anomaly and every material change, and it is written to be read by an investigator on the worst day, not by a marketer on the best one.

EXECUTIVE LENS

Bounds live outside the model, in code the model cannot edit.

MISSION LENS

An accepted residual risk has a name attached, always.

OPERATIONS LENS

Write the safety case for the investigation, not the launch party.

Orbit is governed: filings, licences and liability shape the strategy

Space activity runs under national licensing built on the Outer Space Treaty framework: states authorise and supervise their operators and bear international liability for damage. In practice a space AI strategy must satisfy, as applicable, launch and operations licensing, spectrum filings through the ITU process, remote-sensing licences for Earth imaging, debris-mitigation and end-of-life rules, and export controls on technology and data.

- **Autonomy is disclosable.** Licence applications and renewals describe autonomous functions honestly; an undisclosed capability discovered after an incident is a regulatory failure on top of a technical one.
- **Rules differ by regime:** LEO, MEO, GEO and cislunar operations carry different coordination, disposal and reporting duties; the register records which apply per spacecraft.
- **Standards are evidence:** debris-mitigation guidelines, space-safety standards and insurer requirements form the audit trail a licence leans on.
- **Change triggers notification:** material autonomy changes are assessed against licence conditions before flight, not reported after.

This chapter summarises the landscape for planning; it is not legal advice, and each programme takes advice for its licensing state and regime.

EXECUTIVE LENS

Undisclosed autonomy is a liability multiplier.

MISSION LENS

Map duties per spacecraft and regime in the register.

OPERATIONS LENS

Insurers are an audience for the safety case, not an afterthought.

Command links are crown jewels; AI widens both defence and attack surface

- **Command authentication is absolute:** encrypted, authenticated commanding with keys managed as mission-critical assets; AI systems hold no standing command authority beyond their envelope and session.
- **Jamming and spoofing are operating conditions,** tested in exercises: navigation and timing signals can be degraded or falsified, and autonomy must detect inconsistency rather than follow a confident lie.
- **Supply chain reaches orbit:** flight and ground software provenance, signed builds and vendor access are audited, because an upload pipeline is a command path.
- **AI-specific threats:** poisoned training telemetry, model extraction through query access, and adversarial inputs to screening models are assessed per function.
- **Incident response spans two worlds:** a cyber incident on the ground can become a flight incident on the next pass; playbooks join them, with authority pre-delegated for disconnecting AI from the loop.

EXECUTIVE LENS

Disconnecting the AI is a rehearsed manoeuvre, not an improvisation.

MISSION LENS

A spoofed signal followed confidently is worse than no signal.

OPERATIONS LENS

Vendor access to the upload path is command access. Govern it so.

Rent capability, never the authority to fly

Space AI vendors range from flight-software houses to cloud analytics firms discovering gravity. Procurement tests: flight heritage per function, evidence at the rung claimed, behaviour in degraded modes, data rights over your telemetry and the models trained on it, and exit - can the fleet run, safely and legally, the day the contract ends?

- **Your telemetry trains your models.** Vendor rights to reuse fleet data across customers are priced, bounded or refused explicitly.
- **Sovereignty is operational:** licensing states may require national control of commanding; architectures respect that as design, not paperwork.
- **Escrow and continuity:** models, configurations and documentation needed to operate safely are escrowed or delivered, tested at contract signature, not at dispute.
- **Claims audit:** every vendor performance claim enters the register with its rung and its evidence, and marketing numbers never migrate into safety cases.

EXECUTIVE LENS

Exit is a flight-safety property. Test it early.

MISSION LENS

Data rights over telemetry decide who owns the learning.

OPERATIONS LENS

Heritage is per function; a docking vendor is not a scheduling vendor.

Cost per verified mission outcome, or the savings are fiction

Space AI economics fail in a familiar way: vendor savings counted in operator hours, while the costs land in verification, simulation, licensing, insurance and the one anomaly nobody priced. The unit of this series is **cost per verified mission outcome**: a delivered data product, a completed observation, a safely served customer-year - fully loaded with assurance costs, divided by outcomes telemetry confirms.

- **Three buckets, always separated:** hours claimed, hours released (rosters actually change), hours banked (budget actually falls or capacity is re-deployed to named work).
- **Downside is priced, not footnoted:** expected-loss arithmetic includes spacecraft loss, service interruption and debris liability at honest probabilities.
- **Propellant is money:** manoeuvre-happy autonomy spends mission life; the economics count delta-v as a cost line.
- **Insurance and licensing effects are real savings or real costs:** a strong assurance case can move premiums; an incident moves them further, the other way.

EXECUTIVE LENS

An hour saved that no roster feels is a press release.

MISSION LENS

Count delta-v. Autonomy that burns freely is spending the mission.

OPERATIONS LENS

Price the loss case in the same table as the saving.

One register, one flight authority, anyone can halt



Figure 2 - The amber authorisation ring: autonomy acts inside a human-signed envelope, and the ring, not the model, defines the boundary.

- **The autonomy register** lists every AI function per spacecraft: level, envelope, evidence rung, expiry, owner and restrictions. Unregistered means unauthorised, and discovery of an unregistered function suspends it pending review.
- **Flight authority:** a named flight director (or mission equivalent) owns the authority ladder, promotions and demotions, and can halt any autonomous function without appeal in the moment; review follows, authority first.
- **Anyone can halt:** any operator who sees an autonomous action they cannot explain can stop it and call the pass; halts are logged, thanked and analysed, never punished.
- **Anomaly reviews feed the register:** every anomaly touching an AI function closes with a register action - re-evidence, demote, restrict or retire.
- **Re-licensing is calendar-driven:** licences expire on evidence age, not on confidence; renewal re-runs the gate evidence at current fidelity.

EXECUTIVE LENS

Unregistered means unauthorised, on discovery, every time.

MISSION LENS

Halt authority that needs permission is decoration.

OPERATIONS LENS

Expiry by evidence age keeps the whole fleet honest.

WORKED EXAMPLE - AURORA ORBITAL (ILLUSTRATIVE)

A vendor constellation claim, audited down to flight truth

Aurora Orbital is fictional. All figures are illustrative and labelled as such wherever they are used.

Aurora operates a 24-satellite Earth-observation constellation in sun-synchronous orbit. Its autonomy vendor claimed a 44 percent operations saving and 52,000 operator hours a year across pass planning, telemetry triage, onboard screening and conjunction response, with the fleet described as rung 4, flown, as a whole.

Measure	Vendor claim	Audited (illustrative)
Operations saving	44%	17% verified
Operator hours	52,000 claimed	21,300 released; 13,900 banked
Pass planning	Rung 4	Rung 4 confirmed, A3 on 24 of 24 spacecraft
Telemetry triage	Rung 4	Rung 4 for alerting; missed one slow battery fade class
Onboard screening	Rung 4	Rung 3: flown on 9 spacecraft, sampling audit incomplete
Conjunction response	Rung 4	Rung 2: simulated only; two high-interest events handled manually

The audit found the familiar pattern: the strongest functions were real, the fleet-wide language was not, and the riskiest function carried the thinnest evidence. Nothing in the finding required stopping the programme; everything in it required licensing it honestly.

EXECUTIVE LENS

Audit the claim per function; fleet language hides weak cells.

MISSION LENS

Released and banked are the only hours that fund anything.

OPERATIONS LENS

The riskiest function had the best slides and the least evidence.

WORKED EXAMPLE - DECISIONS (ILLUSTRATIVE)

Licensed in part, refused in part, and the refusals are the system working



Figure 3 - The Aurora operations wall (illustrative): the fleet licensed cell by cell, with the conjunction cell held in red.

Licensed at SS4

- Pass planning at A3 across the fleet, with the independent command checker retained and its override log reviewed weekly.
- Telemetry triage at A2 alerting, with the battery-fade class added to the labelled anomaly library and a re-test within 60 days.
- Onboard screening at A3 on the 9 flown spacecraft, sampling audit raised to 8 percent of rejected scenes until miss rates settle.

Refused or deferred

- Autonomous conjunction manoeuvre execution: refused at any level; AI proposes, flight director decides, coordination log mandatory.
- Fleet-wide screening rollout to the remaining 15 spacecraft in one upload: deferred to a three-spacecraft cohort with rollback rehearsed.
- Vendor reuse of Aurora telemetry to train models for other operators: refused as bundled; a separate priced data agreement may be negotiated.

Cost per verified mission outcome in the licensed cells: 0.83 against a 0.97 pre-AI baseline (illustrative). SS5 scale was granted in part: planning and triage scale; screening scales by cohort; conjunction autonomy does not scale because it was never licensed.

EXECUTIVE LENS

Grant in part is a success state, not a compromise.

MISSION LENS

A refused bundle can return as a priced, separate decision.

OPERATIONS LENS

Scale only what was licensed; the rest is a new application.

THE SPACE STANDARD

The one-page standard every programme in this series signs

- Autonomy is licensed per spacecraft, per orbit, per pass and per failure mode.
- Every function sits on a named rung of the claim ladder, with evidence and an expiry.
- Authority levels A1 to A4 are earned per function; demotion triggers are automatic.
- Hard safety bounds live outside any model, in code no model can edit.
- Every command load passes an independent checker dumber than its author.
- Onboard functions fly frozen, checksummed, with a rehearsed rollback.
- Conjunction decisions are human, coordinated and logged.
- Raw telemetry is never overwritten; gaps are labelled, never silently filled.
- Cost is counted per verified mission outcome, in claimed, released and banked hours.
- Anyone can halt any autonomous function; halts are thanked and analysed.
- Unregistered autonomy is unauthorised autonomy, suspended on discovery.
- Material changes re-open the licence before flight, not after.

EXECUTIVE LENS

The standard fits one page so it can survive one bad night.

MISSION LENS

If a practice cannot be stated simply, it will not be followed at 3 a.m.

OPERATIONS LENS

Sign it at programme level; enforce it at pass level.

FAILURE PATTERNS

Six ways space AI programmes lose spacecraft, money or trust

#	Pattern	Signature
1	Simulation drift	Twin and flight diverge quietly; licences rest on a fiction
2	Fleet-language laundering	One strong spacecraft's evidence sold as a fleet property
3	Alert fatigue	Operators mute the channel that later carries the real anomaly
4	Envelope creep	Thresholds widened by small operational favours, never re-signed
5	Upload optimism	New model flown fleet-wide with an unrehearsed rollback
6	Silent authority	A system acts, telemetry arrives late, and reporting fills the gap with confidence

Each pattern is detectable in the register before it is visible in orbit: stale reconciliation, rung claims without per-spacecraft evidence, alert budgets blown, envelope change logs, upload plans, and reporting that outruns telemetry. The quarterly review in the operating model reads for exactly these six.

EXECUTIVE LENS

Patterns are register findings before they are flight findings.

MISSION LENS

Envelope creep is the slow version of an unauthorised burn.

OPERATIONS LENS

Fatigue is measured, or it is destiny.

ASSURANCE QUESTIONS

30 questions a board, insurer or regulator will eventually ask

- Which functions are autonomous, per spacecraft?
- What rung is each claim on, and what is the evidence?
- Who signed each envelope, and when does it expire?
- What are the automatic demotion triggers?
- Where do hard bounds live, and who can edit them?
- How are command loads independently checked?
- What is the rollback for the last three uploads?
- How many safe-mode entries this year, and why?
- What is the sim-to-flight reconciliation delta?
- Which anomaly classes has triage missed?
- What is the alert budget, and is it met?
- Who decides conjunction manoeuvres, on what thresholds?
- How are other operators coordinated with?
- What happens on a missed pass, a spoofed signal?
- Which licences and filings does autonomy touch?
- What changed since the last licence review?
- Who can halt autonomy, and when did they last practise?
- What telemetry trains which models, under what rights?
- Can the fleet operate if the vendor exits tomorrow?
- What is cost per verified mission outcome, fully loaded?
- How much delta-v did autonomy spend this quarter?
- What is the expected-loss figure, and who accepted it?
- Which functions are refused, and where is that written?
- How is disposal compliance protected from optimisation?
- What is the debris posture of every spacecraft?
- How are keys and command authentication managed?
- When were jamming and spoofing last exercised?
- What evidence is stale, and what is its renewal date?
- Which unregistered functions were found this year?
- What would make you ground the whole system?

EXECUTIVE LENS

If a question surprises the programme, it was the right question.

MISSION LENS

Answers cite the register, not recollection.

OPERATIONS LENS

Question 30 gets a written answer before it gets a live test.

THE GATES

SS0-SS5: the strategy gates for space AI

Gate	Question	Exit evidence
SS0 Commit	Is the problem worth orbital risk at all?	Decision brief, refusal list, named flight authority
SS1 Map	Where does AI pay, stage by stage?	Graded space map, data readiness per stage
SS2 Prove	Does it hold in simulation and shadow mode?	Twin grading, Monte Carlo, shadow-flight results
SS3 Authorise	Is the envelope signed, bounded and expiring?	Safety case, envelope, demotion triggers, rollback
SS4 Fly	Has it operated in flight inside the envelope?	Telemetry-verified performance, reconciliation delta
SS5 Scale	May it extend - per spacecraft, per regime?	Per-cell re-evidence, economics, regulator posture

A gate is passed per function and per spacecraft, with a signer and a date, or it is not passed. Partial passage is recorded as partial, exactly as in the worked example, and scale at SS5 inherits every restriction and the earliest expiry of the cells beneath it.

EXECUTIVE LENS

Gates are per cell; averages do not pass gates.

MISSION LENS

Partial is a respectable verdict. False complete is not.

OPERATIONS LENS

SS5 inherits restrictions - scale never launders them.

FIRST 90 DAYS

A reset plan for a programme starting or correcting course

- **Days 1-15 - Inventory.** Build the autonomy register from flight and ground configurations, not from vendor lists. Suspend nothing yet; mark every unverified claim at its honest rung.
- **Days 16-30 - Refusals and authority.** Sign the standing refusal list, name the flight authority, and rehearse a halt and a demotion on a live exercise.
- **Days 31-50 - Evidence.** Grade the twin per phenomenon, reconcile the last quarter of autonomous actions against telemetry, and label the anomaly library's gaps.
- **Days 51-70 - Economics.** Restate vendor savings into claimed, released and banked; publish cost per verified mission outcome for the two strongest functions.
- **Days 71-90 - Gate the fleet.** Run SS gates per function per spacecraft, license what passes, defer what does not, and set every expiry date in the register.

EXECUTIVE LENS

Inventory from configurations, not brochures.

MISSION LENS

Rehearse the halt before you need the halt.

OPERATIONS LENS

Day 90 ends with expiries set, not just licences granted.

What proves it, and how the regimes differ

The evidence pack per function

Safety case and hazard log; envelope and signatures; twin grading and reconciliation history; shadow and flight performance with misses; upload and rollback records; alert budgets; conjunction coordination log where relevant; data rights and training provenance; economics sheet in claimed, released and banked terms.

Sector annex

Regime	Strategic emphasis
LEO constellations	Fleet operations, conjunction load, pass economics, debris posture
GEO	Long life, station-keeping precision, graveyard disposal, relocation coordination
MEO / navigation	Timing integrity, spoofing resistance, service guarantees
Lunar and cislunar	Light-time delay, navigation without GPS, relay dependence
Deep space	Asynchronous operations, pre-signed contingency, years-long evidence horizons
Crewed	Advisory ceilings, abort authority, safety-of-life refusals

EXECUTIVE LENS

One pack per function; a programme pack hides weak cells.

MISSION LENS

Regime changes the risk, so it changes the licence.

OPERATIONS LENS

Deep space buys autonomy with pre-signed contingency.

TEMPLATES 1-4

Fill-in templates - copy, complete, file in the register

Template 1 - Space AI decision brief

Function / spacecraft / regime / decision sought / rung claimed / evidence / downside priced / refusal check / signer.

Template 2 - Autonomy envelope

Function / level A1-A4 / bounds / prohibited actions / abort conditions / demotion triggers / expiry / halt authority.

Template 3 - Claim and evidence register entry

Claim / rung / evidence artefact / evidence date / half-life / renewal date / owner / restrictions inherited.

Template 4 - Upload authorisation

Change / rung evidence / checksum / rollback image / verification passes reserved / anomaly watch period / signer.

EXECUTIVE LENS

A template completed in pencil beats a platform demo.

MISSION LENS

File templates in the register, where gates can find them.

OPERATIONS LENS

Expiry is a field, not a hope.

Fill-in templates - operations and assurance

Template 5 - Conjunction decision log

Event / data quality / probability and thresholds / options costed in delta-v / other operator contacted / decision / decider / outcome.

Template 6 - Anomaly review

Event / AI involvement / detection path / misses / register action: re-evidence, demote, restrict, retire / library entry added.

Template 7 - Vendor claims audit

Claim / rung asserted / rung evidenced / hours claimed, released, banked / data rights / exit test result.

Template 8 - Gate sign-off SS0-SS5

Function / spacecraft / gate / evidence pack reference / restrictions / earliest expiry / signer / date / next review.

EXECUTIVE LENS

The conjunction log is written for the other operator's investigator too.

MISSION LENS

Every anomaly closes with a register action, or it is not closed.

OPERATIONS LENS

Gate sign-offs name restrictions, not just approvals.

GOLDEN RULES

20 golden rules for space AI

- 1. A modelled manoeuvre is not a flown mission.
- 2. Licence per spacecraft, orbit, pass and failure mode.
- 3. Start where actions reverse.
- 4. Hard bounds live outside the model.
- 5. The checker is dumber than the planner, on purpose.
- 6. Freeze configurations for critical phases.
- 7. Safe mode is a destination, not a disgrace.
- 8. Raw telemetry is never overwritten.
- 9. Gaps are labelled, never silently filled.
- 10. Conjunction decisions are human and coordinated.
- 11. Anyone can halt; halts are thanked.
- 12. Demotion is automatic on its triggers.
- 13. Count delta-v as money.
- 14. Hours count claimed, released and banked, separately.
- 15. Unregistered means unauthorised.
- 16. Evidence expires; confidence does not renew it.
- 17. An upload without rollback is a hope.
- 18. Your burn is someone else's conjunction.
- 19. Rent capability, never flight authority.
- 20. Write the safety case for the investigation.

EXECUTIVE LENS

Rules are short so they survive contact with a real pass.

MISSION LENS

Rule 15 has no exceptions and no grandfathering.

OPERATIONS LENS

Post them in the control room, not the slide deck.

SERIES INDEX

AI in Space and Orbital Operations - the ten volumes

Vol	Title	Gates	Status
1	Space Strategy	SS0-SS5	**Built - this volume**
2	Mission Design and Autonomy	MD0-MD5	Next
3	Satellite and Constellation Operations	SO0-SO5	Planned
4	Ground Systems and Mission Control	GS0-GS5	Planned
5	Earth Observation and Space Data	EO0-EO5	Planned
6	Navigation, Timing and Debris Avoidance	NT0-NT5	Planned
7	Launch, Rendezvous and Servicing	LS0-LS5	Planned
8	Lunar and Deep-Space Operations	LD0-LD5	Planned
9	Space Safety, Security and Assurance	SA0-SA5	Planned
10	Space Capstone and Assurance	CP0-CP5	Planned

Volume 1 sets the strategy and the licensing rule for the series: per spacecraft, per orbit, per pass and per failure mode. Volume 2, Mission Design and Autonomy, takes the rule into design trades, autonomy architectures and the envelopes each mission class can honestly carry.

EXECUTIVE LENS

Ten volumes, one register, one licensing rule.

MISSION LENS

Strategy first, because orbit forgives nothing.

OPERATIONS LENS

Volume 2 begins where this one signs off.